

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

03

NONDETERMINISTIC FINITE AUTOMATA (NFA)

DENNIS GUNAWAN

REVIEW

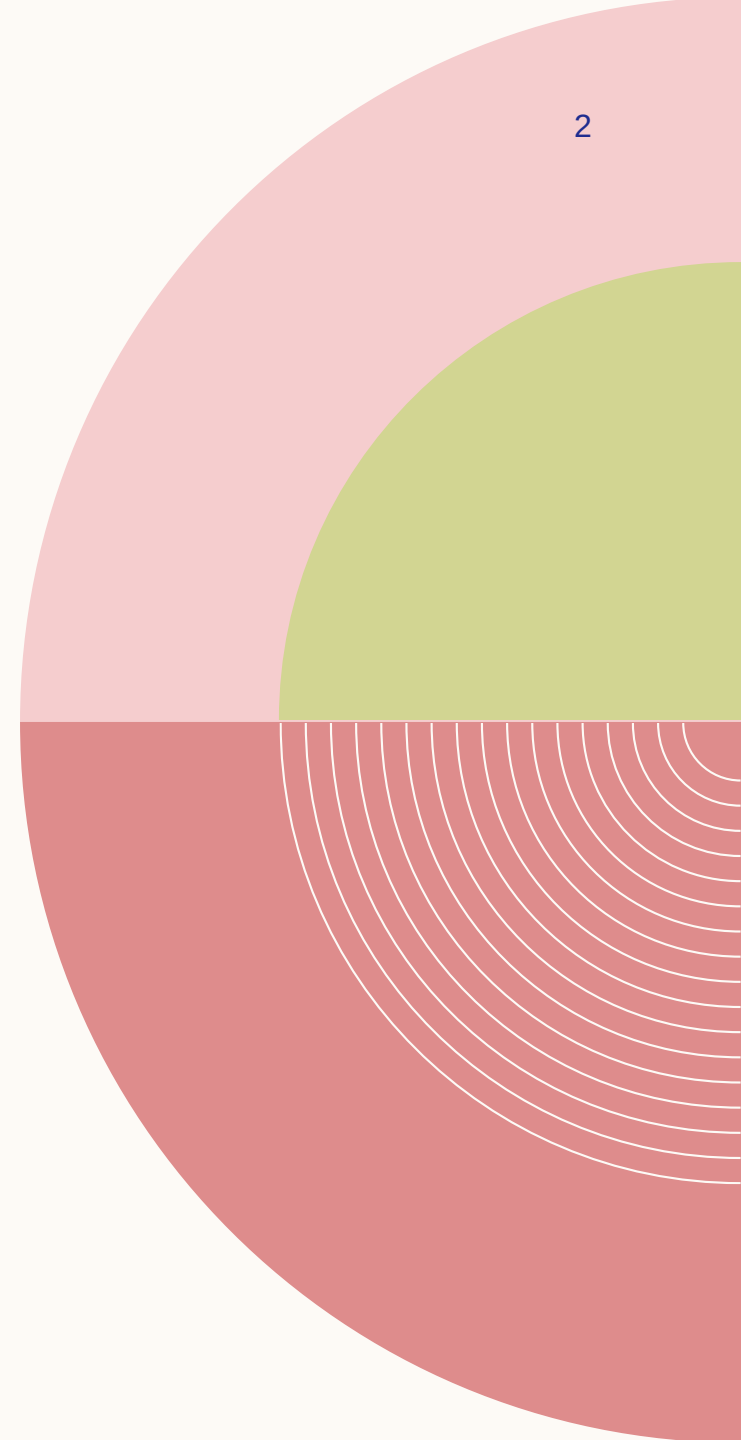
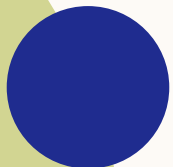
Deterministic Finite Automata (DFA):

Finite State Automata (FSA)

Deterministic Finite Automata (DFA)

Regular Languages

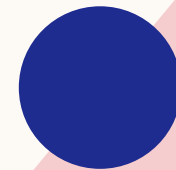
Equivalence of DFA



OUTLINE

Nondeterministic Finite Automata (NFA)

Conversion of NFA to DFA

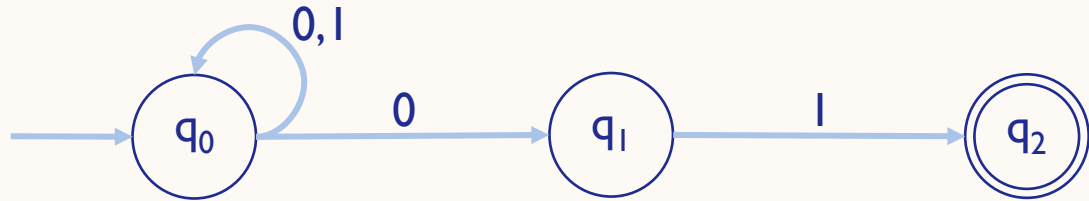


NONDETERMINISTIC FINITE AUTOMATA (NFA)

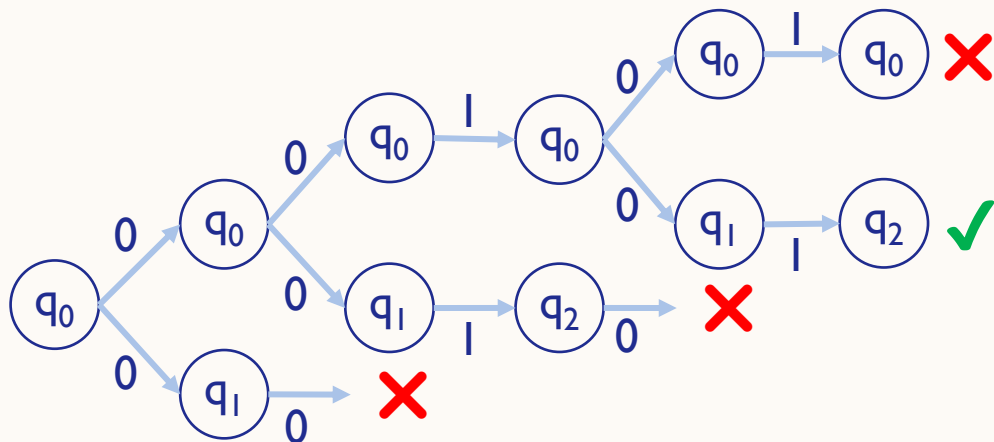
- Nondeterministic Finite Automaton has the power to be in several states at once
- Five-tuple notation: $M = (Q, \Sigma, \delta, q_0, F)$
 - Q : a set of states
 - Σ : a set of input symbols
 - δ : a transition function from $Q \times \Sigma \rightarrow 2^Q$
 - The only difference between an NFA and a DFA is in the type of value that δ returns
 - A set of states in the case of an NFA and a single state in the case of a DFA
 - q_0 : a start state
 - F : a set of final or accepting states

EXAMPLE

- An NFA accepting all strings that end in 01



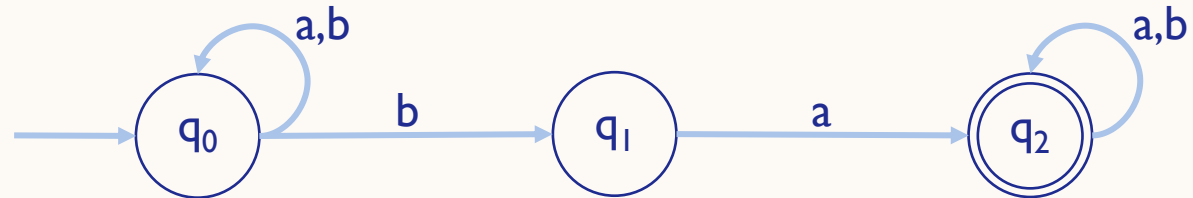
- The states an NFA is in during the processing of input sequence 00101



If there is any way to run the machine that ends in any set of states out of which at least one state is a final state, then the NFA accepts

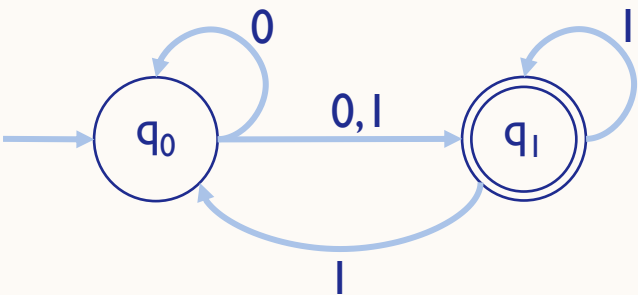
EXAMPLE

Construct an NFA that accepts sets of all strings over $\{a,b\}$ that contain 'ba'.



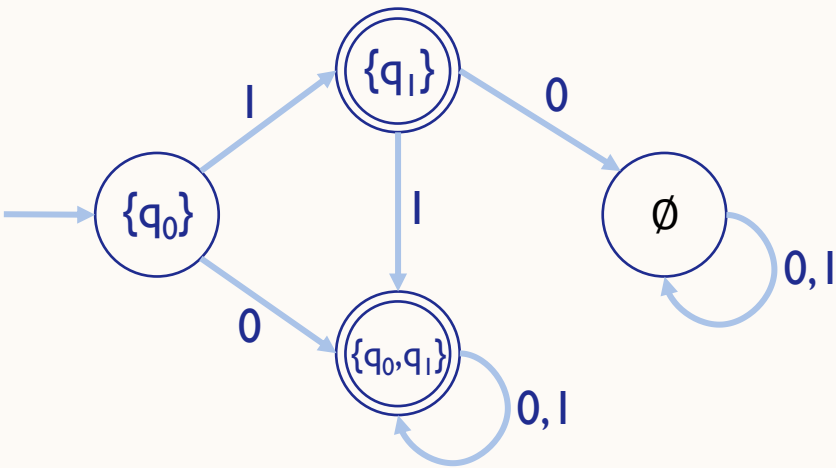
CONVERSION OF NFA TO DFA

Construct the equivalent DFA for the NFA given below



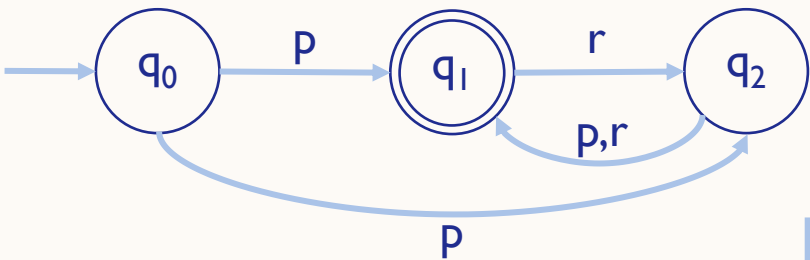
δ	0	1
q_0	$\{q_0, q_1\}$	$\{q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$ $\{q_0, q_1\} \cup \emptyset$	$\{q_0, q_1\}$ $\{q_1\} \cup \{q_0, q_1\}$
q_1	$\emptyset$	$\{q_0, q_1\}$
$\emptyset$	$\emptyset$	$\emptyset$

δ	0	1
q_0	$\{q_0, q_1\}$	$\{q_1\}$
q_1	$\emptyset$	$\{q_0, q_1\}$



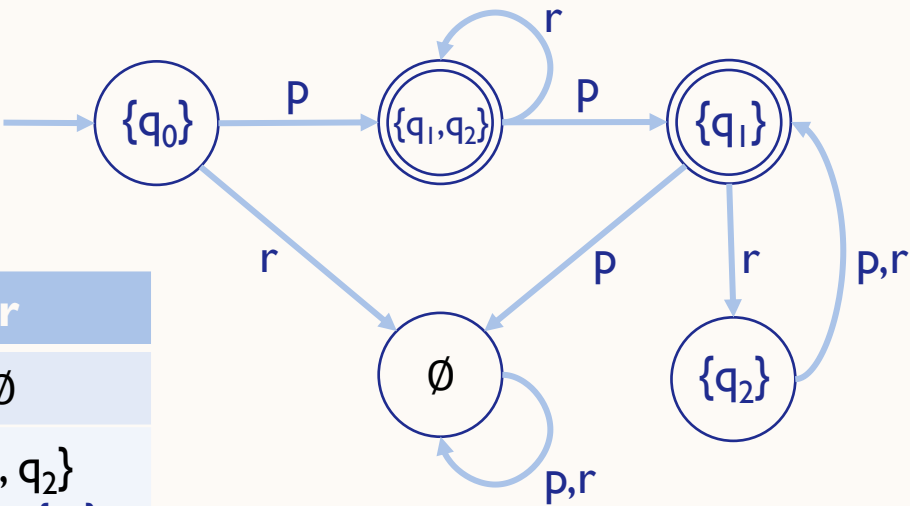
CONVERSION OF NFA TO DFA

Construct the equivalent DFA for the NFA given below



δ	p	r
q_0	$\{q_1, q_2\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\{q_1\}$	$\{q_1\}$

δ	p	r
q_0	$\{q_1, q_2\}$	$\emptyset$
$\{q_1, q_2\}$	$\{q_1\}$ $\emptyset \cup \{q_1\}$	$\{q_1, q_2\}$ $\{q_2\} \cup \{q_1\}$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\{q_1\}$	$\{q_1\}$
$\emptyset$	$\emptyset$	$\emptyset$





PRACTICE

EXERCISES

1. Give the transition diagram for the NFA below.

$$\begin{aligned} Q &= \{q_0, q_1, q_2, q_3, q_4\} \\ \Sigma &= \{0, 1\} \\ q_0 &= q_0 \\ F &= \{q_2, q_4\} \end{aligned}$$

Accepted or rejected?

- a. 001
- b. 10010
- c. 111000

δ	0	1
q_0	$\{q_0, q_3\}$	$\{q_0, q_1\}$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$
q_3	$\{q_4\}$	$\emptyset$
q_4	$\{q_4\}$	$\{q_4\}$

EXERCISES

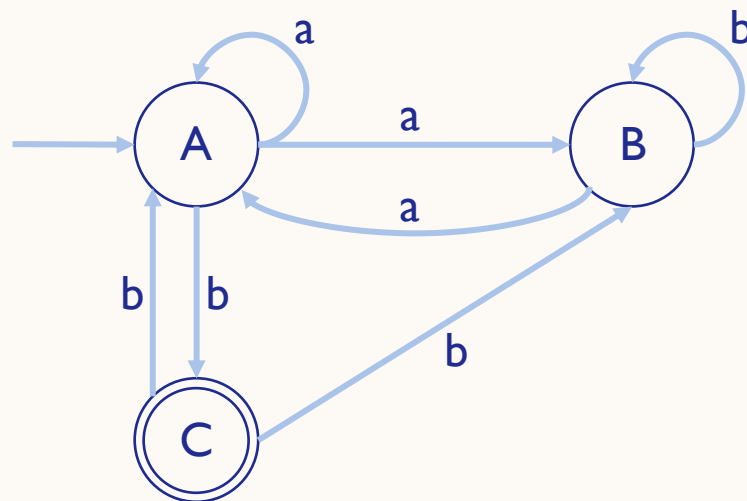
2. Design an NFA accepting set of all strings over $\{a, b\}$ of length 2.
3. Construct an NFA that accepts set of all strings over $\{a, b\}$ that starts with 'ba'.
4. Construct the equivalent DFA for the NFA given below.

$$\begin{aligned}Q &= \{p, q, r, s\} \\ \Sigma &= \{0, 1\} \\ q_0 &= p \\ F &= \{s\}\end{aligned}$$

δ	0	1
p	p, q	p
q	r	r
r	s	$\emptyset$
s	s	s

EXERCISES

5. Construct the equivalent DFA for the NFA given below.



EXERCISES

6. Construct the equivalent DFA for the NFA given below.

$$\begin{aligned} Q &= \{q_0, q_1, q_2\} \\ \Sigma &= \{0, 1\} \\ q_0 &= q_0 \\ F &= \{q_0\} \end{aligned}$$

δ	0	1
q_0	q_1	q_2
q_1	$\emptyset$	q_0, q_2
q_2	q_0, q_2	q_0

7. Design an NFA for a language that accepts all strings over $\{0, 1\}$ in which the second last symbol is always '1'. Then convert it to its equivalent DFA.

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

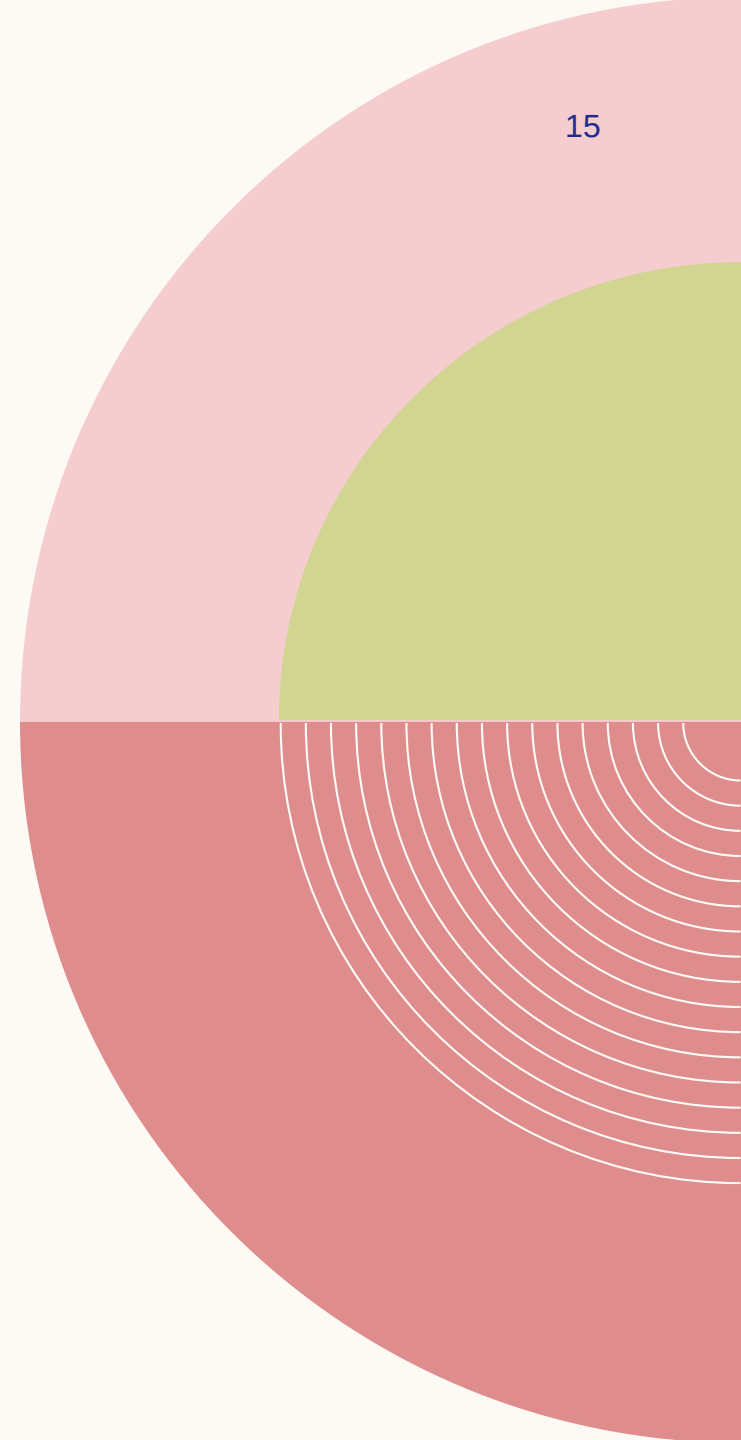
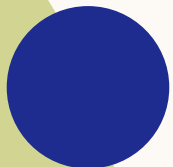
Epsilon-NFA (ϵ -NFA):

Epsilon-NFA (ϵ -NFA)

Epsilon-Closures

Conversion of ϵ -NFA to NFA

Union and Concatenation of FSA



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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